

Quotient Manifolds

1.1 Quotients of Manifolds by Group Actions

1.2 Homogeneous Spaces

Definition 1.2.1

A smooth manifold endowed with a transitive smooth action by a Lie group G is called a **homogeneous G -space**.

Theorem 1.2.2 ► Homogeneous Space Construction Theorem

Let G be a Lie group and let H be a closed subgroup of G . The left coset space G/H is a topological manifold of dimension equal to $\dim G - \dim H$, and has a unique smooth structure such that the quotient map $\pi : G \rightarrow G/H$ is a smooth submersion. The left action of G on G/H given by

$$g_1 \cdot (g_2 H) = (g_1 g_2) H \quad (21.3)$$

turns G/H into a homogeneous G -space.

Theorem 1.2.3 ► Homogeneous Space Characterization Theorem

Let G be a Lie group, let M be a homogeneous G -space, and let p be any point of M . The isotropy group G_p is a closed subgroup of G , and the map $F : G/G_p \rightarrow M$ defined by $F(gG_p) = g \cdot p$ is an equivariant diffeomorphism.

1.3 Quotient Lie Groups